

Abandoned Object Identification and Detection System for Railways in India

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Abstract—Video surveillance systems used in places such as airports, train stations or different public spaces, can bring security to an upper level. The video surveillance systems that are used for security reasons require more intellectual and more robust technical directives. The proposed work compares different methods to detect abandoned baggage in a public area as well as introduces an abandoned luggage detection system for places like railway stations in India. With worries about worldwide security on the ascent, it has turned out to be crucial to have set up proficient risk recognition frameworks that can identify and perceive possibly risky circumstances, and alert the authorities to make a suitable move. This work depicts a framework that perceives the event of somebody leaving baggage unattended in railway stations either intentionally or mistakenly. As tracking of intentionally left baggage is a major problem as it imposes serious security risks in countries like India which has world's largest railway system. Also, proper tracking of mistakenly left baggage is another issue to resolve. The proposed system invokes an alarm whenever it encounters a left baggage as well as generates information about the baggage: its color, date and time when the baggage was left.

Keywords—Video surveillance; Object detection; Crowded environment; Background subtraction.

I. INTRODUCTION

Intelligent and automatic security surveillance systems have recently become a point of focus because of the constantly developing need for such frameworks. In a crowded environment like railway platforms, it gets really tough for the human operator to detect left baggage manually. The intentionally left baggage can lead to severe security threats especially in countries where a large railway system exists as e.g., India has the largest railway system worldwide. Second aspect is keeping track of baggage which are left by mistake. To manually keep track of left baggage is therefore a very time consuming and tiresome task and requires huge manpower to handle. Typically, there is a small team that keeps an eye on a set of displays that cycles different views of various places simultaneously and it decreases the efficiency and creates a threat. To achieve early detection of these threats with the help of automated security frameworks, the capacity to dependably recognize suspicious things and distinguish their owners is desperately essential in public places, for example, railway platforms, airplane terminals and prepare stations. Past studies have given different meanings of an object abandonment event. This study takes after three comparative

yet marginally unique principles. Contextual rule: baggage is viewed as unguarded after the individual who entered the region possessing that baggage in question is no longer around it. Spatial rule: baggage will be viewed as unguarded when its owner is beyond a little neighborhood about the baggage. Temporal rule: when the owner of a baggage leaves the region without the baggage, or upon the baggage being left unattended for few continuous seconds, the baggage is viewed as abandoned or unattended.

This work proposes a framework that perceives the event of somebody leaving baggage unattended in railway stations either intentionally or mistakenly and invokes an alarm whenever it encounters a left baggage. The system generates information about the baggage: its color, date and time it was left unattended. Our directive includes the acknowledgement of four sub-occasions that describe the movement of intrigue. At the point when a lone bag is identified, the framework dissects its past to decide its probable claimer(s); the owner is characterized as the individual who brought the bag into the frame before deserting it unguarded.

II. LITERATURE REVIEW

As of late, much work has been done towards planning frameworks that can detect abandoned objects in open regions. Haritaoglu et. al. depicted a technique that endeavors intermittent movement and static symmetry of a man to decide whether the individual is carrying an object. This Framework utilized adaptive background subtraction (ABS) strategies to find obscure, altered, or evacuated things/objects. Spengler and Schiele suggested a direction for finding out surrendered objects and following individuals by using Condensation algorithm. Martinez-del Rincon and Aguilera have investigated the utilization of multiple cameras in various settings.

The task of object detection is innately difficult because of the extensive variety of articles which could be carried by a man, along with the different routes. It is then made difficult to fabricate a detector for carried objects(s) in view of its guise in solitude or together with the person carrying those objects. Another alternate way is to search for anomalies in the outline of a person to identify whether he/she is carrying some object(s) e.g., baggage. "Haritaoglu et. al proposed a backpack technique to detect objects by observing the person's outline is symmetric when a bag is not being carried.

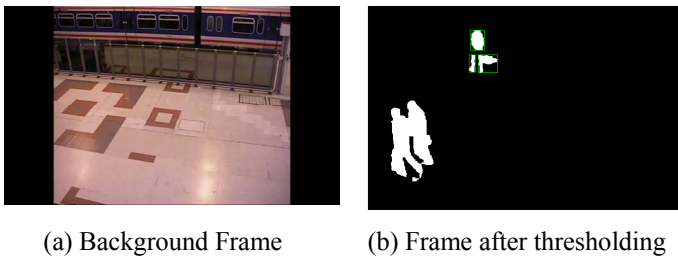


Fig.1. Background and thresholded frames

The method expected individuals to show occasional movement. Further works by Benabdelkader and Davis supplemented the contribution of Haritaoglu and isolated a man's body on a level plane into 3 cuts. The periodicity and the abundance of time arrangement along each cut is concentrated to identify deviations from "normal" strolling individual and found out vertical position of carried objects. They confirmed that the fundamental impediment in Haritaoglu's strategy is the affectability of symmetry to the area and the size of the carried object. Nanda et al. identified pedestrians/walkers carrying things/ objects as outliers of a model for an unhampered person on foot acquired in a managed learning methodology. They used a three layer artificial neural network to classify accordingly. The contribution of Tao et al. tries to recognize people on foot, passing on, by carrying out walk analysis through the use of General Tensor Discriminant analysis (GTDA), they used USF HumanID walk investigation dataset to carry out their study. Ghanem and Davis handles identifying deserted luggage by contrasting the transient layout of the individual before moving toward a Region of Interest (ROI) where object is left. Carried objects are distinguished by contrasting the transient formats and shading histograms of the "earlier" and "after" groupings. The system utilized two static cameras to record comparative perspectives. Later, Motion based recognition approach was used in the detection of objects. Carried items can be categorized into two classes (1.) those which modify the manner in which the individual strolls (that is the biomechanics of step) & (2.) those that modify the way the individual shows up on the ground, that is they impede some portion of the body when carried.

There are two ways to deal with visual identification of a carried item: either decide whether the individual's stride is inside the ordinary range (accepting we have a model of 'typical walk'), or portray the adjustments in look/appearance (related to its shape or surface) which are characteristic of the nearness of a carried protest. The non- parametric approach can only identify a carried thing/object that obtrudes adequately beyond the body profile. Object detection was also done using homographic and simple heuristics. iOmniscient claims to provide intelligent video surveillance software that spots objects left in busy or crowded areas, using a Non-Motion Detection (NMD) technique. Khan and Shah put forward a planar homograph for determining impediments & distinguish the areas of individuals on the ground plane. Much work has been done in this setting with various systems and procedures each having its own particular upsides and downsides. Table 2

presents a summary of different methods proposed by different authors for the detection of abandoned objects.

III. PROPOSED MODEL

Fig. 2. Depicts the major steps of proposed model to identify left baggage. The system comprises of three major steps:

- Background subtraction and static region identification
- Object sort discovery (deserted or removed)
- Abandoned object recognition and information extraction

The system essentially requires a planning calculation (a threshold time duration) to be used to recognize if the baggage is left adequately long to trigger the alert. Our system uses a mixture of Gaussians to analyze and model the presence of moving objects, left items (characterized by static regions), or removed objects (ghosts or false detections). The method starts by extraction of background. Distinctive edges are utilized to get the foreground mask (for moving objects) and the static region mask (for immovable objects). Texture information was utilized to eliminate shadows and to make the calculation running for rapid lighting alterations. For static region mask, a division strategy is created to identify the kind of the static region, essentially beating past edge-based procedures. Just those left articles that meet the defined criteria will invoke the alert. The point when an unaccompanied baggage is identified, the framework analyses its history to decide its owner. The owner is characterized as the individual who carried the baggage on the platform before leaving it unguarded.

Figure 2 shows the flow of Object detection framework. The system first performs background extraction followed by thresholding step to detect object regions. Fig. 1 shows background frame used for background subtraction. The next step is to separate region of left baggage from the human object. Finally the left baggage properties of interest are extracted as e.g. its color, date and time of leaving it.

The data set used for the study is performance evaluation of tracking and surveillance (PETS) 2006 data set to validate the proposed system.

The proposed method is highly viable as it can detect an abandoned object and extract the features such as the time, date and color of the bag.

Static Region Detection: When an object or a body shows no change for a specific period of time it is said to be Static region. There are many algorithms to detect these static regions. Gradient based technique uses static foreground technique for the detection of static regions by investigating the adjustment in edge vitality connected with edges of static frontal area district of current frame and background frame. According to this technique the object is abandoned if there are comparatively more edges but this technique fails in the frame with multiple objects.

In the proposed system static region detection is based on background subtraction, where every frame is compared with

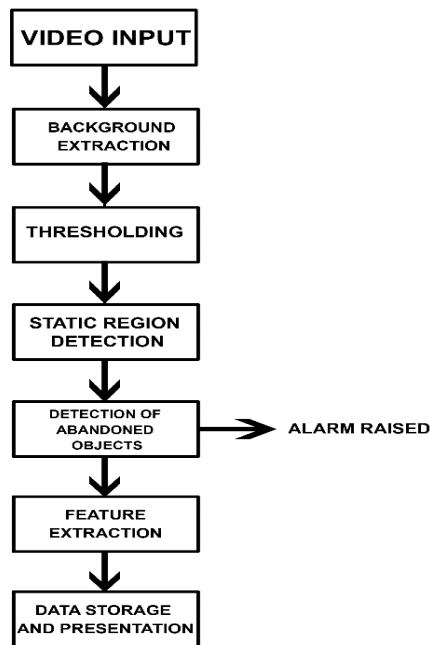


Fig. 2. Flow diagram of proposed system showing different steps

the background frame and static regions are detected. Relying on the scenario, the transitory static areas demonstrate abandoned objects. In other words, an abandoned object can be considered as temporary static object which was not in the frame before. Hence, we use two frames to compare with background image to assure the motion of the object. The time duration for considering an object to be static, changes from environment to environment. With current data set, the static time window for every abandoned object was found to be 85-100 frames. Any object which was fixed at a location for 85-100 frames was considered to be a static one.

Detection of abandoned object: After an object is detected the method creates a box around the object for a specific period of time and raises an alarm after the specified time. A screenshot is popped up with the frame number and the count of detected objects. The box color changes to red until the bag remains unattended. If the owner of the bag reaches the bag over a sequence of successive frames and takes the bag away then the alarm gets deactivated. This choice is made more than a few edges to add to its certainty. Caution is activated and endures till the owner comes back to take the left baggage.

Fig. 3 shows an abandoned luggage detected by the proposed system.

Feature Extraction: The proposed system also helps in extracting the relevant features of the unattended objects i.e. baggage. These feature are color of the bag, date and time of the event and the frame number of the live stream. These information is then stored in a database and also presented to human supervisor and a screenshot of the live stream is taken for the better association with the owner. An alarm is triggered after the detection of any abandoned object and persists till the owner returns or it is physically reset by the human supervisor. This alarm triggers after the owner leaves the baggage for a

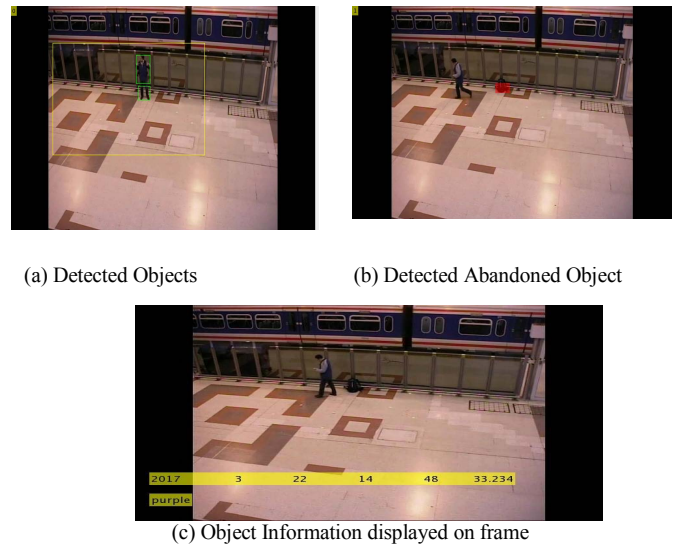


Fig. 3. Intermediate steps are shown (a) – (b) final result in (c)

specific period of time which is easy to tune in the system. The system stores the relevant details of the abandoned baggage in .csv format. It also stores the screenshots taken at the time of the detection.

IV. RESULT AND DISCUSSION

We tested the method on PETS 2006 Dataset and following results were observed as presented in Table I. The method was tested on seven different conditions with different types of baggage eg. Trunks, bags, suitcases and multiple owners. The scenarios contains different difficulties based on close proximity to the item of luggage. The method successfully detected abandoned objects in most of the conditions. The static time window for every abandoned object was taken to be 85-100 frames. Any object which was fixed at a location for 85-100 frames was considered to be a static one and further. The system can accommodate a maximum of 100 object profiles for simultaneous decision making and further processing. The system also considers only a single camera view for detecting left objects. The system was tested on different shots taken from PETS 2006 data set.

TABLE I. RESULTS

Shots	Alarm Count	Abandoned Object Detected	No. of Objects Detected
<i>Dataset S1</i>	85	Yes	1
<i>Dataset S2</i>	85	Yes	1
<i>Dataset S3</i>	85	Yes	1
<i>Dataset S4</i>	95	Yes	1
<i>Dataset S6</i>	85	Yes	1
<i>Dataset S7</i>	85	No	-

TABLE II. SUMMARY OF LITRATURE

Methods	Method used	Advantages	Limitations	Data Set	Accuracy
The Backpack system.	combination of occasional movement estimation with static symmetry	Power and applicability	Objects displaces the principle axis resulting in lesser efficiency.	Not Available	Not Available
Motion-based Recognition Approach	Track and detect person in initial frames then classifies as baggage tagging based on spatiotemporal observations	Vigorous to division and tracking blunders, because takes shape over many edges	The person should be walking in upright pose with a constant speed for some seconds.”	Not Available	85 %
Using homographies and simple heuristics	Background subtraction to detect object followed by a heuristic method for tracing and detection of objects	Simple and flexible	camera frame-work should be aligned onto a ground plane	PETS 2006	Not Available
Creating Temporal Template	Markov arbitrary field along with a map of earlier probabilities and spatial congruity presumption	Powerful than disruptive single frame segmentations	Low detection from distending dress or non-normal form	PETS 2006	Not Available
Tracking Using Region-Matching	Utilizing different synchronized cameras situated at considerable distance	Tracks objects in a jumbled scenario	doesn't indicate change in coordinates from past spans	Not Available	Not Available
Tracking using Geometric Shape Models	couples patio-temporal consistency, defining the event, with standard properties	Simple and efficient	Excludes occurrences such as give, drop, pick up etc.	PETS2006 and MINDSEYE2012	Not Available
Using Planar Homography Constraint	Spatio-temporal areas are grouped using Graph Cuts and each group represents a track of a man and a cut in time	Precisely track individuals in all perspectives	Requires no calibration	Not Available	Not Available
Capture & Exploit Temporal Flow of Events	When an unaccompanied pack is identified, the framework examines its history to decide its possessor(s)	Straightforward & theoretically stable. Can track simultaneous objects	Suffres with False negatives	i-LIDS dataset	Not Available
Optical Flow Motion Model	Speed and course independent movement descriptor	Suffers from lighting alterations and color contrast	Requires no calibration	PETS2006	Not Available
Pose Preserving Dynamic Shape Models	Models dynamic shape disfigurements using kinematics complex embedding	Continuous location of anomaly, and revision of noise outline by gap filling & anomaly deletion	Does not test gait recognition with carrying items/objects	CMU Mobo data set.	Not Available
Lagrangian Dynamics	Reinterprets Lagrangian highlights in connection to underlying movement process	Avoids the removal of unequivocal edge formations by applying HOG descriptor	Not effective for repeating motion Events	Pets2006 dataset	Not Available

V. CONCLUSION AND FUTURE SCOPE

The work proposes a framework that performs object identification and feature extraction of the unattended objects which are left purposely or mistakenly in a crowded environment. The proposed system is engaging in its straightforwardness and reasonably stable. The proposed system utilizes a static frame as a base frame to extract object

information from frames which is a simple and easy approach yet may act as a limiting factor under similar but different scenarios. A future expansion of this work will be to incorporate background estimation and extraction of relationship of the objects with the owner and picking up the details and identity of the owner.

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